

DL for ENG

LECTURE 8.1

Stationary Heat Transfer

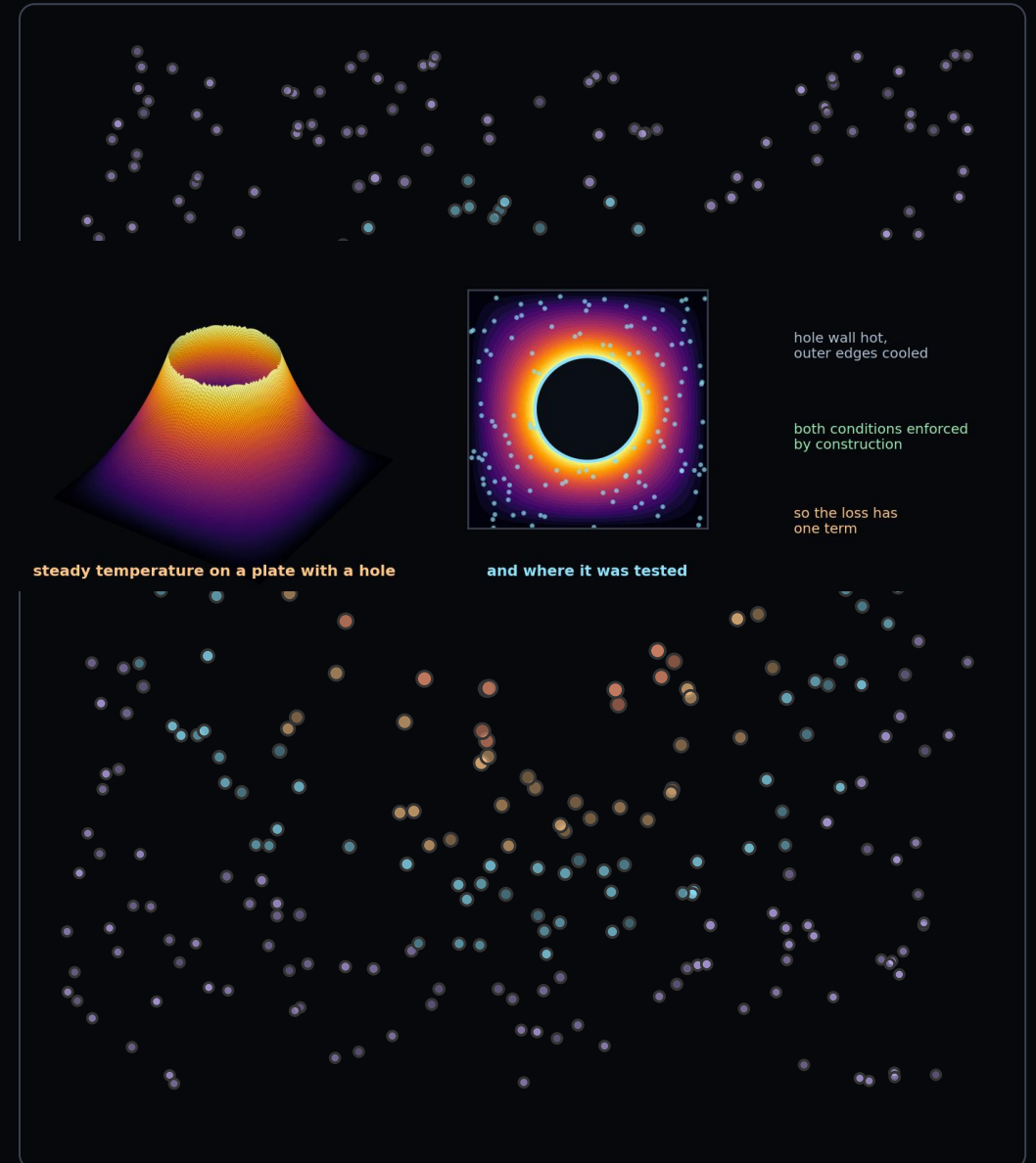
From the elliptic problem of L7.2 to conduction with real units, real boundary conditions and real geometry

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Liu (2025), PINN with Python, Ch. 2.4 and Ch. 3



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **write** — the conservative form, and say why expanding it is wrong at an interface
- **recognise** — a pure-Neumann problem, and say what it needs to be well posed
- **scale** — a device problem before training, and say what breaks if you do not
- **hard-enforce** — a fixed-temperature face, and estimate what that buys
- **handle** — a curved boundary: the normal, the sampling, and the sign
- **identify** — an unknown conductivity from a few thermocouples

WHAT TODAY ASSUMES

from L7.1	residual, composite loss, hard enforcement
from L7.2	Poisson, and its boundary conditions
from L7.1	the T3 quadrature machinery

THE PHYSICS IS REAL NOW

Everything until here was a manufactured problem on a unit square. From this slide the units, the geometry and the material data are a device.

AND ONE NEW CONDITION

Robin cannot be hard-enforced. It is the first condition that must stay soft.

What Changes Now That the Physics Is Real

CARRIED OVER UNCHANGED

- ✓ The network is the temperature field, $T \approx \mathbb{N}\hat{w}(x, y)$.
- ✓ Autograd supplies ∇T and $\nabla^2 T$ exactly.
- ✓ The loss is squared residuals, PDE plus conditions.
- ✓ Hard enforcement still beats penalties.
- ✓ Adam explores, L-BFGS converges.
- ✓ Sampling still obeys $N_{\text{sample}} \geq P^*$.

NEW IN AN APPLICATION LECTURE

- Physical units — and therefore scaling problems.
- A third boundary condition: convection (Robin).
- Geometry that is not a rectangle.
- Material properties that may be unknown.
- An engineering question behind every solve.
- A tolerance that comes from the application.

The engineering question behind L8.1: given a device that dissipates a known power, how hot does it get and where? Everything in this lecture serves that question — which is also why we start with the steady state, the temperature a device settles at when it has been running long enough.

L8.2 then asks how long it takes to get there, and what happens under a changing load.

From Fourier's Law to Poisson's Equation

FOURIER'S LAW

$$\mathbf{q} = -k \nabla T$$

Heat flows down the temperature gradient, at a rate set by the conductivity k .

ENERGY BALANCE

$$\nabla \cdot \mathbf{q} = Q \quad \Rightarrow \quad -\nabla \cdot (k \nabla T) = Q$$

In steady state, what flows out of a control volume equals what is generated inside it.

CONSTANT k

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{Q}{k} = 0$$

For a homogeneous material k comes out of the divergence, leaving Poisson's equation.

This is the same operator you met as Poisson in L7.2 — but Q/k now has units of K/m^2 , and that changes how you must scale the problem before training.

When the Conductivity Varies

THE GENERAL FORM — k INSIDE THE DIVERGENCE

$$\nabla \cdot (k(\mathbf{x}) \nabla T) + Q(\mathbf{x}) = 0$$

- **k is rarely uniform** — aluminium bonded to ceramic, a pack of different cells
- **do not expand the divergence when k varies** — that form is wrong at an interface
- a PINN handles the conservative form directly, on the graph
- **across an interface k jumps and the gradient jumps with it** — the normal flux does not

TWO MATERIALS, ONE INTERFACE



T is continuous, its slope is not

Three Thermal Boundary Conditions

DIRICHLET — FIXED TEMPERATURE

$$T(\mathbf{x}) = \tilde{T}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega_D$$

A surface held at a known temperature: a baseplate on a coolant channel, a contact with a large thermal mass.

NEUMANN — PRESCRIBED FLUX

$$-k \frac{\partial T}{\partial n} = \bar{q}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega_N$$

A known heat flow per unit area. The special case $\bar{q} = 0$ is an insulated or symmetric face, and is extremely common.

ROBIN — CONVECTION

$$-k \frac{\partial T}{\partial n} = h(T - T_\infty)$$

The flux depends on the unknown temperature itself: Newton cooling into ambient air at T_∞ with film coefficient h .

Robin is the one that is new relative to L7.2 — and it is the one that actually describes a device sitting in air. The next slide explains why it needs care.

Why the Convection Condition Is Harder

ROBIN RESIDUAL — BOTH A VALUE AND A DERIVATIVE OF THE UNKNOWN

$$\mathcal{L}_R = \frac{1}{n_R} \sum_{i=1}^{n_R} \left[-k \frac{\partial T}{\partial n} - h(T - T_\infty) \right]^2$$

- Dirichlet fixes T, Neumann fixes the flux, Robin fixes a combination
- and Robin cannot be hard-enforced by a simple multiplier
- so it is a soft term by necessity, balanced against the residual
- the Biot number tells you how much that matters

BIOT NUMBER

$$Bi = \frac{hL}{k}$$

PRACTICAL GUIDANCE

Compute Bi before you train. If it is very large or very small, replace Robin with the limiting condition — you will get the same answer from a better-conditioned loss.

Well-Posedness and the Pure-Neumann Trap

COMPATIBILITY CONDITION FOR A PURE-NEUMANN PROBLEM

$$\oint_{\partial\Omega} \bar{q} \, ds = \int_{\Omega} Q \, d\Omega$$

- if every boundary carries a flux and none fixes a temperature, T is determined only up to a constant
- and a solution exists only if the fluxes balance the generation exactly
- **a PINN will not warn you** — it converges to some member of the family
- two fixes: pin the temperature at one point, or constrain the mean

THE SAME FIELD, SHIFTED

all three satisfy the PDE and every flux condition

Scaling: Do It Before You Train

CHANGE OF VARIABLES

$$\theta = \frac{T - T_0}{T_1 - T_0}, \quad \xi = \frac{x}{L}, \quad \eta = \frac{y}{L}$$

THE EQUATION KEEPS ITS FORM

$$\frac{\partial^2 \theta}{\partial \xi^2} + \frac{\partial^2 \theta}{\partial \eta^2} + \widehat{Q} = 0$$

- a silicon device has **L ≈ 1 mm, k ≈ 150, Q ≈ 1e8** — raw numbers span many orders
- scale length by the device size and temperature by the physical rise
- every trial solution in this lecture assumes it
- **train scaled, present physical** — invert the transformation at the end

A SILICON POWER DEVICE

QUANTITY	PHYSICAL	SCALED
length L	1.0e-3 m	1
conductivity k	150 W/mK	1
source Q	1.0e8 W/m³	0.67
temperature rise	0 → 45 K	0 → 1
film coeff. h	25 W/m²K	Bi = 1.7e-4

Bi ≪ 1 here: the surface is nearly insulated, so the convection condition could be replaced by ∂T/∂n = 0 with negligible error.

Residual and Composite Loss

RESIDUAL

$$\mathcal{F}_{\widehat{w}} = T_{,xx} + T_{,yy} + \frac{Q}{k}$$

COMPOSITE LOSS

$$\mathcal{L} = \mathcal{L}_{PDE} + w_D \mathcal{L}_{DBC} + w_N \mathcal{L}_{NBC}$$

- both second derivatives come from two autograd passes on one graph
- interior points test the residual; boundary points carry the soft conditions
- hard-enforce the Dirichlet term and it disappears from the loss entirely
- **the source is known in a forward problem** — make it unknown and this becomes inverse

$\mathcal{L}_p^d$	interior points	<i>always present</i>	$\mathcal{L}^{DBc}$	fixed-temperature faces	<i>removable by trial solution</i>
$\mathcal{L}^{NBc}$	flux and insulated faces	<i>soft; may need weighting</i>	$\mathcal{L}^R$	convective faces	<i>soft by necessity</i>

Hard-Enforcing a Fixed-Temperature Face

TRIAL SOLUTION FOR A PRESCRIBED TEMPERATURE ON ALL FOUR EDGES

$$T_{trial}(x, y) = \bar{T} + x(1-x)y(1-y) \mathcal{N}_{\hat{w}}(x, y)$$

- the same construction as L7.1, now reading as physics
- the network supplies the interior; the multiplier holds the face
- for a non-uniform prescribed temperature, evaluate it on the boundary
- **measured in L7.1** — about 165× more accurate in relative L2
- and the held face is usually the interface you care about

WHICH FACES CAN BE HARD-ENFORCED

Fixed temperature	yes — multiplier
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Insulated, $\bar{q} = 0$	sometimes
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Prescribed flux	no
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Convection (Robin)	no
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A mixed problem therefore always retains at least one soft term. Hard enforcement reduces the balancing problem; it rarely eliminates it.

Weighting the Flux Terms

LIU'S DOCUMENTED EXCEPTION

$$w_N = 100$$

- Liu's default is one for every term, tuned only when a term is ignored
- **the reason is dimensional** — a flux residual and a PDE residual carry different units
- non-dimensionalising first removes most of it
- **diagnose before you tune** — plot the per-term losses separately

PER-TERM LOSS HISTORY

log₁₀ loss. The flat amber curve is the failure: the total is falling, but the flux condition is not being learned.

Sizing the Network and the Sample Set

PSEUDO-DIMENSION

$$P^* = (p + 1) + (N_n + 1) N_L$$

- P^* counts neurons, not weights
- sample density should follow the temperature gradient, not the geometry
- concentrate points near sources, cooled faces and geometric features
- **a starting point** — ten to thirty times P^* interior points

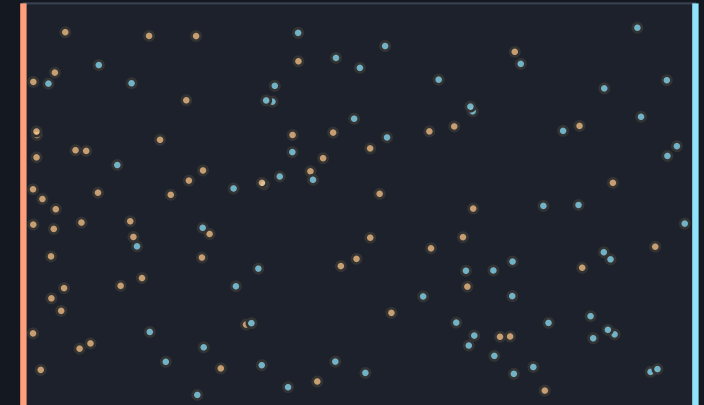
SAMPLING CONDITION

$$N_{sample} \geq P^* \quad N_{sample} \gg P^*$$

DENSITY FOLLOWS THE GRADIENT

heat source

cooled face



denser sampling where the gradient is steep

Why Rectangles Are Not Enough

- every hard-enforcement trick so far has assumed a rectangle
- real thermal hardware is not rectangular
- **and geometry is not cosmetic** — a hole beside a source concentrates flux
- two things break: the multiplier has no closed form, and uniform sampling misses the feature

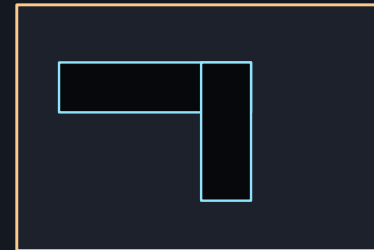
GEOMETRIES THAT ACTUALLY OCCUR



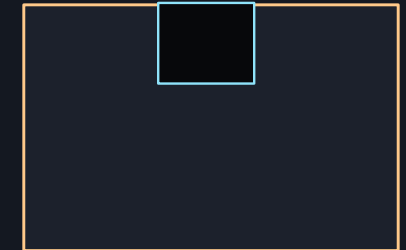
plate with a hole



fin array



channel with a bend



tab / notch

Triangulating the Domain

ELEMENT AREA

$$A = \frac{1}{2} |\mathbf{u} \times \mathbf{v}| = \frac{1}{2} |u_x v_y - u_y v_x|$$

NODAL WEIGHT

$$w_i = \sum_{T \ni i} \frac{A_T}{3}$$

QUADRATURE

$$\int_{\Omega} g \, d\Omega \approx \sum_{i=1}^n w_i g(\mathbf{x}_i), \quad \sum_{i=1}^n w_i = A_{\Omega}$$

- **the mesh does not hold the solution** — the network still does
- it exists only to place sample points and supply integration weights
- a genuine difference from FEM: no shape functions, no assembly
- **and a coarse mesh suffices** — its job is coverage, not accuracy

T3 MESH AROUND A HOLE



elements shrink towards the hole, where the flux concentrates

A Learned Boundary Multiplier

HARD ENFORCEMENT WITHOUT A CLOSED-FORM MULTIPLIER

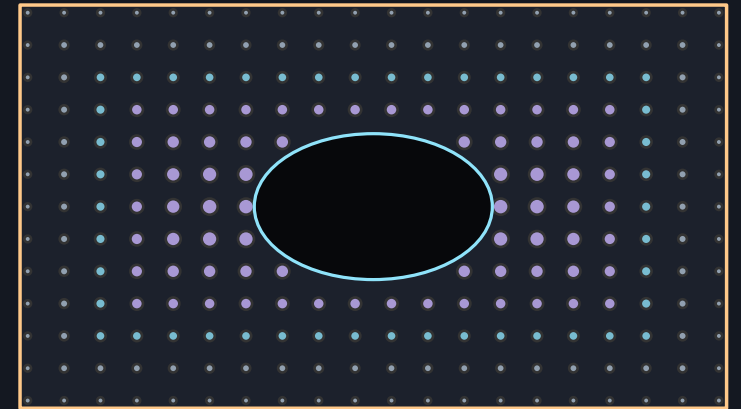
$$u_{trial}(x, y) = \mathcal{N}_{\hat{v}}^{DBC}(x, y) \mathcal{N}_{\hat{w}}(x, y)$$

- on a rectangle the multiplier is a product of linear factors
- on a plate with an elliptical hole there is no such closed form
- so train a small second network to be that function, once, in advance
- the cost is one cheap pre-training run
- for an analytic shape the level set itself will do

ANALYTIC SHORTCUT FOR AN ELLIPSE

$$\left(\frac{x - x_c}{a}\right)^2 + \left(\frac{y - y_c}{b}\right)^2 = 1$$

THE MULTIPLIER FIELD



zero on the outer edge, positive inside

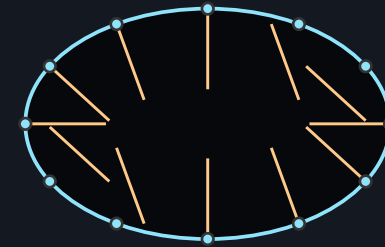
Flux Conditions on a Curved Boundary

THE NORMAL DERIVATIVE, EVALUATED POINTWISE

$$\frac{\partial T}{\partial n} = \nabla T \cdot \mathbf{n}$$

- on a rectangle the normal is one of four constant vectors
- on a curved boundary it varies point to point
- **the recipe is unchanged** — autograd gives the gradient, you supply the normal
- get the sign right: the outward normal of the domain points into the hole
- and sample the curve by arc length, not by angle

NORMALS AROUND A HOLE



n points OUT of the material, into the hole
each sample point carries its own normal

The Energy Route for Steady Conduction

TOTAL POTENTIAL FOR STEADY CONDUCTION

$$\Pi(T) = \frac{1}{2} \int_{\Omega} k \nabla T \cdot \nabla T \, d\Omega - \int_{\Omega} Q T \, d\Omega - \int_{\partial\Omega_N} \bar{q} T \, ds$$

- the steady field minimises this functional
- so the flux condition appears as a boundary work term and needs no loss
- in exchange the fixed-temperature condition becomes essential
- Liu's measured trade** — about 10× faster, typically less accurate

COLLOCATION OR ENERGY?

	COLLOCATION	ENERGY
derivative order	2nd	1st
flux condition	loss term	natural
fixed temperature	either	must be hard
needs quadrature	no	yes
convergence	slower	~10× faster
accuracy	higher	lower

Manufacturing a Solution to Check Against

CHOOSE THE ANSWER FIRST, THEN DERIVE THE SOURCE THAT PRODUCES IT

$$T = \sin(\pi x)\sin(\pi y) \quad \Rightarrow \quad \frac{Q}{k} = 2\pi^2 \sin(\pi x)\sin(\pi y)$$

- pick a field, substitute it in, and read off the source that makes it exact
- now you have a reference to check against
- this is the only way to separate discretisation error from formulation error
- do it before touching real geometry or real material data

PRIMARY METRIC

$$\varepsilon_{L_2} = \frac{\|T_{\hat{w}} - T_{\text{exact}}\|_2}{\|T_{\text{exact}}\|_2}$$

REPORT ALL THREE

Relative L_2 for the overall fit.

Maximum absolute error — the hot spot is what a thermal design is certified on.

Maximum error on each boundary, separately.

Convergence: What Should Fall, and What Should Not

- a PINN has no mesh to refine, so the convergence study is different
- refine the sample set and the error should fall, then plateau
- the plateau tells you the network, not the sampling, is now the limit
- widening raises P^* , so the minimum sample count rises with it
- if nothing improves along any axis, the formulation is wrong

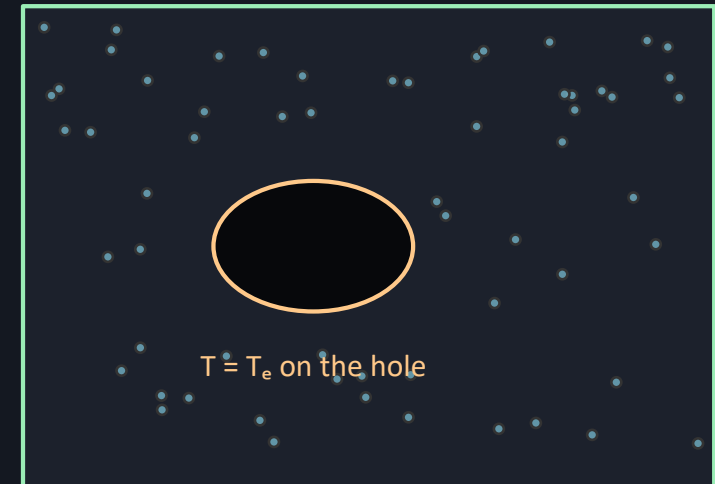
ERROR AGAINST SAMPLE COUNT

$\log_{10}$ relative L_2 against interior point count. Each curve flattens where the network, not the sampling, becomes the limit.

Heated Plate with a Cooling Hole

- the canonical device problem, and the one Liu carries through both chapters
- a plate generating heat, cooled through a hole
- fixed temperature on the hole, insulated outer edges, uniform generation
- the engineering question is the peak temperature and where it sits
- and the same geometry returns in L8.2 with time added

PROBLEM DEFINITION



insulated outer edges ($\bar{q} = 0$)

uniform internal generation Q throughout

the peak sits on the edge furthest from the hole

Building the Model, Step by Step

1 Scale

Map the plate to the unit square and the temperature rise to $[0, 1]$. Compute B_i to check whether convection matters.

2 Mesh

Triangulate the domain with elements refined towards the hole. Extract nodes and nodal weights.

3 Multiplier

Use the hole's level-set function to hard-enforce its fixed temperature. No pre-training needed for an ellipse.

4 Sample

Interior points from the mesh; boundary points by arc length on the hole and uniformly on the outer edges.

5 Assemble

Residual from two autograd passes; add the flux loss on the insulated edges with its own weight.

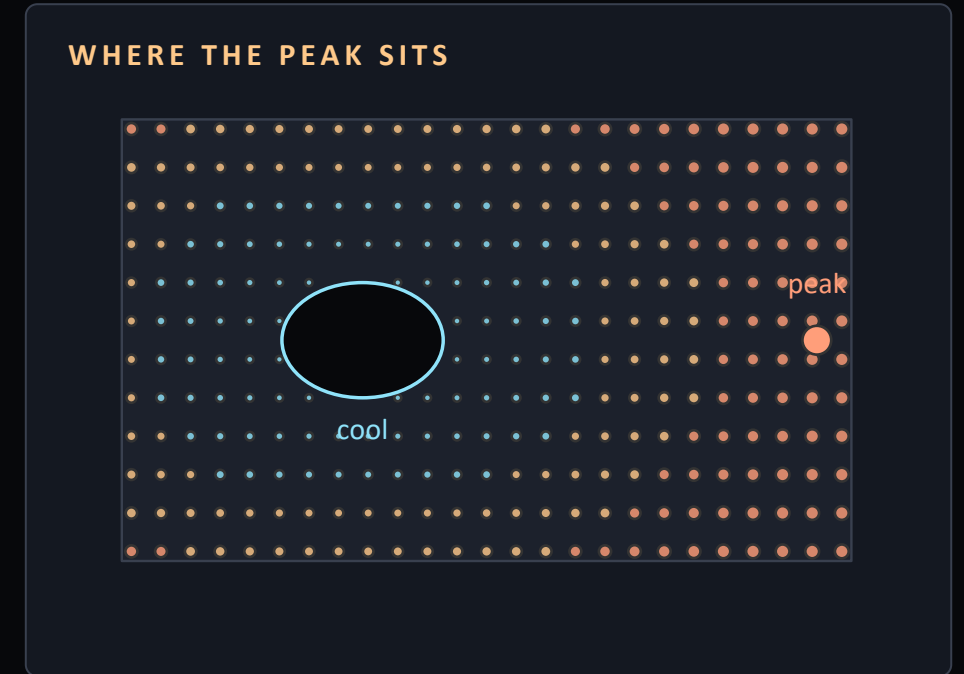
6 Train

Adam to explore, L-BFGS to converge. Watch the per-term losses, not just the total.

Steps 1 and 2 are where most projects go wrong, and neither involves a neural network. Scaling and geometry come before any training.

Reading the Result Like an Engineer

- a **contour plot is not a result** — a number with a tolerance is
- check the flux balance: what leaves through the hole must equal what is generated
- look at the residual map, not only the loss
- a small mean residual can hide a large local one exactly where it matters
- then ask the design question the model was built to answer



● **Peak temperature** against the limit

● **Residual map** no local hot spots

● **Flux balance** in = out, to a few %

● **Boundary error** on each face separately

Identifying an Unknown Conductivity

THE FORWARD MODEL, WITH k PROMOTED TO A TRAINABLE PARAMETER

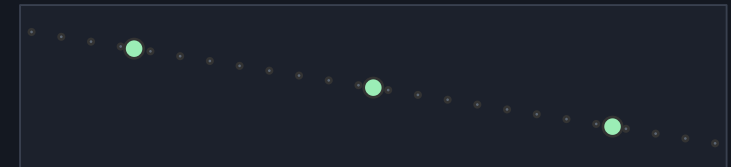
$$k \text{ trainable}, \quad \mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{data}$$

- thermal conductivity is one of the least reliable numbers in a datasheet
- promote k to a trainable parameter and add a data term from a few thermocouples
- the same network, the same loss, one more trainable quantity
- this is the highest-value use of a PINN in thermal work
- **but watch identifiability** — sensors on an isotherm recover nothing

SENSOR PLACEMENT MATTERS



clustered — k poorly identified



spread across the gradient — k recovered

PINN or FEM for a Steady Thermal Problem?

USE FEM

A single forward solve on fixed geometry with known properties.
Sharp material interfaces.
You need a certified answer with an established error bound.
Decades of validation behind it.

USE A PINN

Unknown parameters to identify from sparse sensors.
Many solves across a parameter sweep, where a trained model evaluates instantly.
Geometry that resists meshing.

- ▶ For one steady conduction solve on a meshable domain, FEM is faster, more accurate and better validated. Saying so is not a weakness of the course — it is what lets you recognise the cases where a PINN genuinely wins.
- ▶ Those cases are real and they are the ones this course targets: inverse problems, parametric studies, and coupling with data. All three are awkward for a classical solver and natural here.

The honest position: a PINN is not a better solver. It is a differentiable one — and differentiability is what buys you inverse problems and design sensitivity.

Pitfalls Specific to Stationary Heat

Unscaled physical units

Conductivities of 10^2 and sources of 10^8 fed straight into a tanh network.

Symptom: loss barely moves from its initial value. Fix: non-dimensionalise first.

Pure-Neumann degeneracy

Every face carries a flux condition, so the temperature level is undetermined.

Symptom: perfect residual, wrong absolute temperature. Fix: pin one point.

Flux term ignored

The PDE residual dominates and the insulated faces are never learned.

Symptom: total loss falls, flux term flat. Fix: weight it, as Liu does with $w^N = 100$.

Normal sign error

The outward normal on an internal hole points into the hole, not away from it.

Symptom: heat flowing the wrong way; a plausible but inverted field.

Material interface

A smooth network cannot represent the kink in ∇T at a conductivity jump.

Symptom: smeared interface. Fix: separate networks per material, coupled at the interface.

The first two are formulation errors and no amount of training fixes them. Check both before you blame the optimiser.

Questions

TEN TO TEST WHETHER TODAY LANDED

- why must the divergence stay unexpanded when k varies?
- what does a jump in k do to the gradient, and what stays continuous?
- why can a Robin condition not be hard-enforced?
- what is wrong with a problem whose boundaries are all flux conditions?
- **why scale before training** — and which construction depends on it?
- what does hard enforcement buy on a fixed-temperature face?

AND FOUR MORE

- | | |
|--------------|---|
| seven | why sample a curved boundary by arc length? |
| eight | what does the triangulation do, if not hold the solution? |
| nine | how do you check a steady result without a reference? |
| ten | when would you use FEM instead, and say so plainly? |

THE ONE WITH NO SETTLED ANSWER

Your sensors all sit on one isotherm. Can k be recovered at all?

BEFORE L8.2

Ex_08.1 notebooks 01 to 03. Bring the flux-balance number.

Ex_8.1 — Steady Conduction in a Plate with a Hole

TASKS

- 1 Verify on a manufactured solution: solve on the unit square and reproduce the Ex_7.1 accuracy.
- 2 Add an internal heat source with physical units, then non-dimensionalise and confirm the answers agree.
- 3 Introduce an elliptical hole; sample the domain and hard-enforce its fixed temperature by the level-set multiplier.
- 4 Impose zero flux on the outer edges. Sweep w^N over 1, 10, 100 and plot the per-term losses.
- 5 Report peak temperature, its location, and the flux balance in versus out.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

Why must the problem be scaled before training? sl. 8

Why can a convection condition not be hard-enforced? sl. 6, 10

What breaks if every face carries a flux condition? sl. 7

How do you know the flux condition was actually learned? sl. 11, 22

When would you use FEM instead, and why? sl. 24

Built on the same core library as Ex_7.1 and Ex_7.2 — only the residual, the conditions and the geometry change.

What Time Will Add in L8.2

	L8.1 — STATIONARY	L8.2 — DYNAMIC
The equation	$\nabla^2 T + Q/k = 0$	$\partial T / \partial t = c \nabla^2 T + Q / (\rho c_p)$
Inputs	x, y	x, y, t
Conditions	<i>BCs only</i>	<i>BCs + one initial condition</i>
Domain	Ω	$\Omega \times (0, t_{en}^d]$
New failure mode	—	<i>causality: late times fit before early ones</i>
Engineering question	<i>how hot?</i>	<i>how fast, and under what duty cycle?</i>

Same geometry, same material, same code. One term is added to the operator, and one condition to the loss.

Key Takeaways

- 1 It is Poisson, with units**

Steady conduction is the elliptic problem of L7.2 carrying physical quantities. The mathematics is unchanged; the scaling, the conditions and the geometry are not.
- 2 Scale before you train**

A device with $L = 10^{-3}$ m and $Q = 10^8$ W/m³ will not train unscaled. Non-dimensionalisation is a necessary step, not a tidiness measure.
- 3 Convection resists hard enforcement**

Robin constrains a value and a derivative at once, so no multiplier satisfies it. It stays soft, and therefore needs weighting.
- 4 Geometry is the real work**

Meshing, multipliers and normals take more effort than the PINN itself. The mesh places points and supplies weights — it never holds the solution.
- 5 The win is the inverse problem**

For one forward solve, use FEM. For an unknown conductivity recovered from three thermocouples, use this.

Next — L8.2: Dynamic Heat Transfer. Same plate, same hole, with time.